

Viscosity and Surface-Promoted Slippage of Thin Polymer Films Supported by a Solid Substrate

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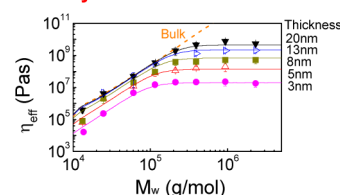
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S Supporting Information

ABSTRACT: Thermally activated flow dynamics of polystyrene films supported by silicon is studied for a wide range of film thickness (h_0) and molecular weights (M_w). At low M_w , the effective viscosity of the nanometer thin films is smaller than the bulk and decreases with decreasing h_0 . This is due to enhancement of the total shear flow by the augmented mobility at the free surface. As M_w increases, with h_0 becoming smaller than the polymer radius of gyration (R_g), the effective viscosity switches from being substrate-independent to substrate-dependent. We propose that interfacial slippage then dominates and leads to plug flow. The friction coefficient is found to increase with h_0 providing $h_0/R_g < \sim 1$, demonstrating a surface-promoted confinement effect.

Effective Viscosity of PS-on-SiOx Nano-films at 120°C



INTRODUCTION

Many recent experiments reveal that polymer under confinement in the nanometer range often exhibits dynamics distinguishably different from the bulk.^{1–5} While the mechanisms underpinning the new properties are still controversial, most results suggest that they originate from surface effects. Usually, the chain dynamics near the free (air) surface is faster than the bulk^{6–11} but that near the substrate is slower.^{12,13} By embracing analogous effects in a layer model, variations in the dynamic properties, including the glass transition temperature, T_g ,^{10,14,15} and effective viscosity, η_{eff} ,^{8,12,16} with the average polymer thickness, h_0 , were explained. At high molecular weights, M_w , where the unperturbed radius of gyration, R_g , exceeds $\sim h_0$, de Gennes further proposed that the high-mobility chain segments at the free surface can bring about mobility enhancement to the whole chain through chain connectivity.¹⁷ Here, we report a comprehensive study on the η_{eff} of polystyrene supported by oxide-coated Si (PS-SiOx; SiOx thickness = 102 ± 5 nm) with $13 \leq M_w \leq 2300$ kg/mol ($\sim 3 < R_g < \sim 41$ nm (Supporting Information)), polydispersity index 1.01–1.1, and h_0/R_g from ~ 0.2 to ~ 10 . A new dynamic regime dominated by plug flow and instigated by a surface-promoted confinement effect is found.

EXPERIMENTAL METHODS

Polystyrene (PS), with a weight-average molecular weight, M_w , of 13.7–2300 kg/mol and polydispersity index of 1.01–1.1, was purchased from Scientific Polymer Products (Ontario, NY). In this experiment, we use as-cast films made from spin-coating. Methods of sample preparation and characterization have been reported before.^{8,18} In brief, the as-purchased polymer granules are first dissolved in toluene, and then the solution is filtered through a PTFE membrane filter with pore size $0.1 \mu\text{m}$ (Fisher Scientific Co.) before being spin-

coated onto a substrate to form the films. The substrates, purchased from Siltronix (France), are single crystal (001) silicon wafers covered by a 102 ± 5 nm thick thermal oxide. Before use, the wafers are cut into $1.5 \text{ cm} \times 1.5 \text{ cm}$ slides and submerged in a piranha solution at 130°C for 10 min, followed by thorough rinsing in deionized water and then drying by 99.99% nitrogen. Afterward, the slides are further cleaned in oxygen plasma for 20 min whereupon they are ready for use. The thickness of the polymer film is controlled by adjusting either the concentration of the polymer solution or spinning speed and monitored by ellipsometry at five different locations on the film. Deviations found in the measured thickness are typically less than 0.3 nm.

Because the films are smoother than equilibrium, they roughen upon heating. The surface topography of the films is measured by using atomic force microscopy (AFM) at various roughening times, t . We limit t to the initial roughening stage where the film roughness is less than 10% of the average film thickness, and no holes are detectable in the films by AFM. These ensure that linear analysis is valid for the data taken. Each topographic data is Fourier transformed and processed to give the power spectral density (PSD), $A_q^2(t)$.⁸ Figure 1 shows a representative sequence of PSDs we obtain. Unlike the unentangled films (where $M_w < M_e \approx 18$ kg/mol¹⁹), the PSDs of this ($M_w = 393$ kg/mol) and other entangled films do not evolve with time at first (except for the rapid jump in the first time step, which is attributable to the glass-to-rubber transition.²⁰) Even though the PSD is stagnant, frame-to-frame comparison of the AFM images taken in an *in situ* measurement shows that the surface topography is not, indicating the films to be in a quasi-steady state, undergoing equilibrium vibrations. We ascribe the vibrations to the normal modes of the films in the rubbery elastic state. At later times, growth of the PSDs commences. As explained below, this evolution is dominated

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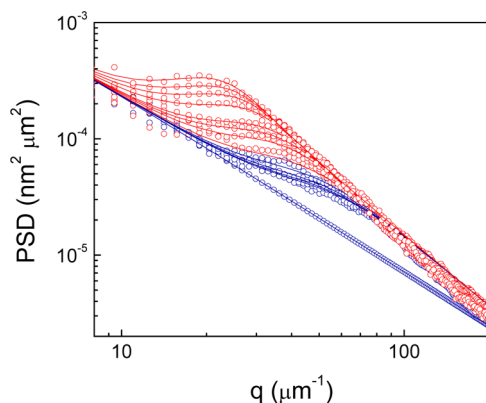


Figure 1. Illustration of the experimental data and model used to determine the effective viscosity, η_{eff} . The symbols represent the power spectral density of a PS-SiOx film with $h_0 = 5$ nm and $M_w = 393$ kg/mol annealed at 120 °C for times. From bottom to top, blue: 0, 600, 1200, 2400, 4800, 9600, and 24000 s; red: 36000, 48000, 96000, 144000, 192000, and 253420 s. The solid lines are model lines with $\eta_{\text{eff}} = 1.5 \times 10^8$ Pa s and $\mu_0 = 11.0$ kPa.

by surface-enhanced shear flow for $M_w < \sim 100$ kg/mol but surface-promoted plug flow at higher M_w 's.

There are thus two dynamic processes with distinct time scales, accountable for the initial stagnant and subsequent growing PSDs for the entangled films. Adopting an adiabatic approximation, we analyzed the slow evolution in the presence of the ensemble-averaged quasi-equilibrium elastic vibrations. The slow evolution is described by the in-plane transport currents, $\mathbf{j}(\mathbf{r}, t)$, caused by fluctuations in the local film profile, $h(\mathbf{r})$, that produces gradients in the local pressure $P(\mathbf{r}, t)$ according to the relation $\mathbf{j}(\mathbf{r}, t) = -M_{\text{tot}} \nabla P(\mathbf{r}, t)$, where M_{tot} is the total mobility of the films.²¹ For the fast vibrations, we assume the mode energy to be $[3\mu_0/(2h_0^3 q^2)]|u_q|^2$, consistent with elastic vibrations with wavevector $\mathbf{q}$ and amplitude u_q ,²² where $q \equiv |\mathbf{q}|$ and μ_0 is the shear modulus of the film. A linear stability calculation assuming lubrication approximation and stable films gives²¹

$$A_q^2(t) = A_q^2(0) \exp(2\Gamma_q' t) + \left(\frac{k_B T}{\gamma_s q^2 + G''(h_0)} \right) (1 - \exp(2\Gamma_q' t)) \quad (1)$$

where $\Gamma_q' = -M_{\text{tot}} q^2 [(\gamma_s q^2 + G''(h_0))^{-1} + (3\mu_0/h_0^3 q^2)^{-1}]^{-1}$. Details of this calculation can be found in ref 21. In eq 1, k_B is the Boltzmann constant, T is absolute temperature, γ_s is surface tension, and $G(h_0)$ is the van der Waals potential of the film. To perceive the physical meaning of this equation, one observes that for cases where $\Gamma_q' < 0$ $\lim_{t \rightarrow \infty} A_q^2(t) = k_B T / [\gamma_s q^2 + G''(h_0)]$. So, eq 1 may be rewritten as $A_q^2(t) = [A_q^2(0) - A_q^2(\infty)] \exp(2\Gamma_q' t) + A_q^2(\infty)$. From this expression, it is apparent that the first term causes $A_q^2(t)$ to change with time whenever $A_q^2(0)$ is different from the equilibrium value. If it is, it evolves to diminish the difference at a rate constant given by Γ_q' . The last term ensures that $A_q^2(t) = A_q^2(0)$ at $t = 0$. Taken together, eq 1 portrays a dynamic process by which the film surface evolves from the initial structure toward the equilibrium one by way of the surface modes with relaxation spectrum given by Γ_q' . The solid lines in Figure 1 display the best fit to eq 1. As seen, the fitted lines agree with the experiment quite well.

If the slow process is caused by viscous shear flow in a film where the viscosity, η , is uniform and there is no slip at the substrate boundary, then $M_{\text{tot}} = h_0^3/(3\eta)$ and our model is identical to the Maxwell model, as demonstrated in ref 21. We thus define the effective viscosity by^{8,21}

$$\eta_{\text{eff}} \equiv h_0^3/[3M_{\text{tot}}] \quad (2)$$

In general, η may be inhomogeneous in the film and slippage may occur. Then M_{tot} may adopt a different form. By modeling M_{tot} or η_{eff}

to match the experiment, one may gain insight about the nature of the slow process.

RESULTS AND DISCUSSION

Figure 2a displays our main result where η_{eff} is plotted versus M_w for PS-SiOx with $3 \leq h_0 \leq 20$ nm. We first focus on the

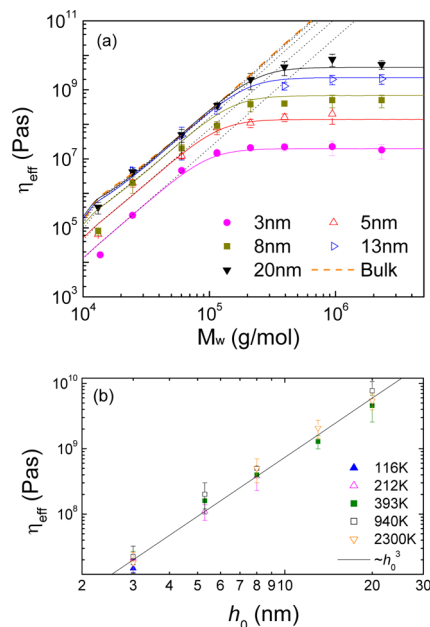


Figure 2. (a) Effective viscosity versus molecular weight from the PS-SiOx films at 120 °C. The film thicknesses, h_0 , are labeled in the figure legend. The dashed line denotes the bulk viscosity. The dotted lines are fits to the two-layer model (eq 3). The solid lines are fits to the model embracing both enhanced surface mobility and surface-promoted interfacial slippage (eq 4). (b) Effective viscosity plotted versus h_0 using the plateau data in (a). The solid line is a straight line with slope 3 to demonstrate that the data empirically follow the $\eta_{\text{eff}} \sim h_0^3$ scaling.

data with $M_w < 100$ kg/mol. There, one finds that η_{eff} agrees with η_{bulk} for the two largest h_0 (i.e., 13 and 20 nm). This validates our method for the viscosity measurement and the assumptions underlying eq 2, namely uniform η and no slip. In practice, the latter should be taken for meaning that any effects due to viscosity inhomogeneity or slippage are insignificant. Empirically, we find that the data of all the films fit well to $\eta_{\text{eff}} \sim M_w^{3.2 \pm 0.2}$, and so do the bulk data (dashed line). It is common to use the M_w dependence of viscosity to corroborate the mechanism underneath the dynamics in polymer.²³ Specifically, for $M_w > M_c$ the characteristic M_w (≈ 31 kg/mol¹⁹), the reptation model prevails and predicts $\eta \sim M_w^{3.4}$,²³ but for $M_w < M_c$ the Rouse $\eta \sim M_w$ prediction is expected although the actual scaling can be complicated by the M_w dependence of segmental friction.²⁴ More importantly, for any given M_w , η_{eff} decreases with h_0 and deviates from the bulk value at sufficiently small h_0 . Previous experiments found that enhanced flow mobility at the free surface produces analogous viscosity reduction in unentangled polymer films.^{8,12} We continue to employ the two-layer model we developed^{8,12} to treat the data in the lower M_w region. In short, by assuming the films to contain a surface mobile layer atop a bulk-like layer and that there is no slippage at the substrate and lateral interfaces, a solution to the Navier–Stokes equation gives^{8,25}

$$M_{\text{tot}} \approx h_0^3 / (3\eta_{\text{bulk}}) + M_{\text{mobile}} \quad (3)$$

where M_{mobile} is the mobility of the surface mobile region. In eq 3, a cross-term⁸ is neglected for simplicity. The estimated error is <20% if the thickness of the mobile layer with enhanced flow transport is 1 nm.²⁶ This is insignificant compared to the size of the error bars. Since a similar M_w dependence is apparent for all the data, we assume the same scaling, $\eta \sim M_w^{3.2 \pm 0.2}$, for the surface mobile region, which amounts to writing $M_{\text{mobile}}^{-1} = a_{\text{mobile}}^{-1} M_w^{3.2}$, with a_{mobile} being a constant. A fit to the data gives $a_{\text{mobile}} = (4.0 \pm 0.5) \times 10^9 \text{ nm}^3 \text{ Pa}^{-1} \text{ s}^{-1} \text{ g}^{3.2} \text{ mol}^{-3.2}$. Expressing this in η_{eff} the fitted result is shown by the dotted lines in Figure 2a. The good fit justifies the two-layer model. We thus ascribe the observations in the $M_w < 100 \text{ kg/mol}$ domain to surface-enhanced shear flow in the film.

Now we contemplate the prospect of this mechanism in the high M_w domain. The relevant factor to consider is the thickness of the surface mobile layer. Because M_{mobile} should be dependent on the total number of mobile chains contained in the surface mobile layer, if the film thickness had been reduced to below the thickness of this layer, M_{mobile} should decrease. The fact that for any given M_w a single value of M_{mobile} is able to fit the experimental η_{eff} down to $h_0 = 3 \text{ nm}$ (dotted lines, Figure 2a) indicates that the thickness of the surface mobile layer is no more than 3 nm and independent of M_w in the lower M_w regime. One may observe that variations exist in the literature for the value of the surface mobile layer thickness.^{7,9,10,27} We surmise that they arise because different experiments probe different aspects of the dynamics, which may have different susceptibility to the phenomenon of enhanced surface dynamics and thereby exhibit different spatial extent from the free surface. More recently, Lang and Simmons²⁸ demonstrated that even variations in how “interfacial dynamics” is defined can alter the surface mobile layer thickness and whether this layer thickness increases or decreases with temperature. It is well established^{29,30} that polymer chains near an impenetrable wall are oblate, with a perpendicular radius of gyration $R_{\perp} \approx 0.5R_g$.³⁰ We estimate that at $M_w \approx 60 \text{ kg/mol}$ $R_{\perp} = 3 \text{ nm}$, namely, the upper limit of the high-mobility layer discussed above. So as M_w increases above $\sim 60 \text{ kg/mol}$, the surface chains should extend into the inner layer, where the slower local segments would slow them down, causing η_{eff} to converge toward the bulk viscosity. This however is not seen in our data as we now discuss.

For $M_w > 100 \text{ kg/mol}$, Figure 2a shows that η_{eff} is depressed even more than surface-enhanced shear flow predicts (dotted lines). More surprisingly, η_{eff} becomes independent of M_w . Measurements obtained from the 3 nm films at various temperatures between $T = 84$ and $143 \text{ }^\circ\text{C}$ (Figure 3) show that the noted M_w -independence of η_{eff} is not unique to $T = 120 \text{ }^\circ\text{C}$. A dramatic change in the M_w dependence in different M_w domains strongly suggests a change in the nature of the slow process. An ostensible M_w independence had been found in the dewetting dynamics of PS supported by liquid-like polydimethylsiloxane³¹ and η_{eff} of PS-Si.²⁰ We do not think that these findings are related to ours. In either case, another slower dynamic process with relaxation time³¹ or η_{eff} ²⁰ displaying the usual $\sim M_w^{3.4}$ scaling was subsequently found when the films were annealed further. But in here, η_{eff} is deduced from the slow process. No additional process could be identified, as shown by the constant η_{eff} found when we annealed the films further until deep holes formed in the film whereupon eq 1 may no longer be valid (Figure S1). Numerous experiments found that as-cast,

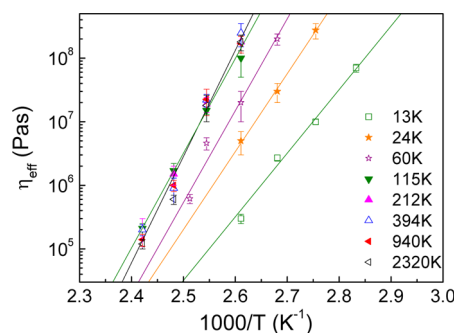


Figure 3. η_{eff} versus reciprocal temperature of 3 nm thick PS-SiOx with various M_w as indicated in the figure legend.

high- M_w films are out-of-equilibrium, exhibiting reduced chain entanglement, residual stress, and irreversible chain adsorption to the substrate.^{31–37} Robustness of the measured η_{eff} in prolonged heating indicates that the noted transitory out-of-equilibrium phenomena bear little influence on our measurements.

We ponder whether a viscosity $\sim M_w^0$ scaling could be consistent with prevailing ideas of polymer dynamics. In the $M_w > 100 \text{ kg/mol}$ domain, all the films in Figure 2a fulfill the $h_0 \leq \sim R_g$ condition, and so the density of chain entanglement can be reduced.³⁸ Should the chains become unentangled, the dynamics would be of the Rouse-type and exhibit $\eta \sim M_w$, not M_w independence.²³ In reptation dynamics, diffusivity of a polymer chain in a tube is $D = k_B T / f$, where f is the total chain friction.²³ The reptation time taken for a chain to diffuse the contour length of the tube L is thus $\tau_{\text{rep}} = L^2 / D \sim f L^2$.²³ On physical grounds, the M_w exponent of f (and obviously that of L , too) should be positive and nonzero for linear polymers. This has indeed been found in the diffusive dynamics of polymer in bulk²³ and also under a variety of confinement and boundary conditions.^{39–42} Therefore, the viscosity arising, being $\sim \tau_{\text{rep}}$,²³ cannot be M_w independent. Correspondingly, any form of reptation-like dynamics involving translation of individual chains past each other must not be responsible for the $\eta_{\text{eff}} \sim M_w^0$ observation. A natural choice for the slow process is therefore plug flow.

This proposed explanation implies that η_{eff} should change when the films are deposited on a different kind of substrate. We study the η_{eff} of PS-HSi (hydrogen-terminated Si, it is more strongly adsorbed by PS than SiOx.¹⁸) Shown in Figure 4 is the

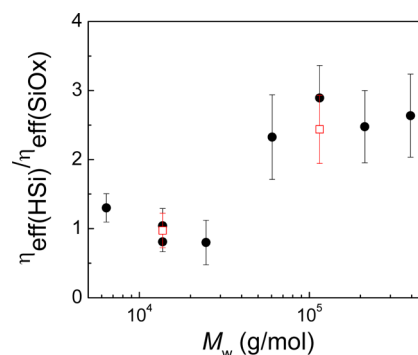


Figure 4. Ratio of η_{eff} from 3 and 5 nm PS-HSi to PS-SiOx versus M_w . For the 3 nm films (circles), the data with M_w below (above) 13.2 kg/mol were taken at 90 $^\circ\text{C}$ (120 $^\circ\text{C}$). The data of the 5 nm films (squares) were taken at 120 $^\circ\text{C}$.

η_{eff} ratio of 3 and 5 nm PS-HSi to 3 nm PS-SiOx, plotted versus M_w . The ratio is close to one at low M_w , in keeping with the presumption that enhanced surface flow dominates the flow dynamics in the lower M_w regime. But as M_w is increased above ~ 60 kg/mol, the ratio switches to ~ 2.5 , showing that the new substrate now admits more resistance to the flow. This is expected if slippage becomes dominant and the polymer–substrate attraction in the new substrate is stronger. The transition point of ~ 60 kg/mol also coincides with the M_w where η_{eff} begins to depart from $\sim M_w^{3.2 \pm 0.2}$ and cross over to $\sim M_w^0$. All these support the proposed dominance of plug flow in $M_w > 100$ kg/mol.

On the basis of the different slow processes discussed, we generalize eq 3 to

$$M_{\text{tot}} \approx h_0^3/(3\eta_{\text{bulk}}) + M_{\text{mobile}} + h_0^2/\xi \quad (4)$$

where the third added term is known to account for slippage⁴³ and ξ is the polymer–substrate friction coefficient.⁴³ For homogeneous fluid with viscosity η , it is related to the slip length, b , by $b = \eta/\xi$.⁴³ For thick films, it is established that ξ is independent of h_0 ⁴⁴ because friction is relatively local to the interface. The added term due to slippage, if dominates, predicts $M_{\text{tot}} \sim h_0^2$, which is consistent with plug flow.⁴³ It is apparent from Figure 2a that the other two terms (accounted for by the dotted line) become negligible when η_{eff} bends into the plateaus. At where the plateaus establish, we find that constancy of ξ breaks down as evident from the empirical scaling $\eta_{\text{eff}} \sim h_0^3$ (equivalently, $M_{\text{tot}} \sim h_0^3$) observed above in the η_{eff} plateau regime (Figure 2b). Dependence of ξ on h_0 indicates there to be another confinement effect.

It is tempting to conclude that $\xi \sim h_0^2$. However, this scaling is apparent only over a limited range of h_0 . We are also unaware of any theoretical basis for such a scaling. We contemplate the following interpretation instead. Interfacial friction often involves microscopic stick–slip motions of occasional “sticky” points with the substrate.⁴⁵ As noted above, the films exhibiting the η_{eff} plateaus have $h_0 \leq R_g$. So the chains involved in the sticky points can extend across the whole films and reach to the top. Enhanced surface dynamics may agitate the sticky points,¹⁷ thereby promoting slippage and reducing friction. From the above, enhanced surface mobility causes a boost in M_{tot} by an enhancement factor of $1 + r$, where $r \equiv M_{\text{mobile}}/[h_0^3/(3\eta_{\text{bulk}})]$. We envisage a similar boost to ξ^{-1} . To allow for variations, we modify the enhancement factor to $1 + \alpha r$, where α should be of order unity. Therefore, $\xi^{-1} = (1 + \alpha r)\xi_0^{-1}$, where ξ_0^{-1} is the value as $h_0 \rightarrow \infty$. The best fit obtained using these assumptions and published η_{bulk} ⁴⁶ is shown by the solid lines in Figure 2a. Evidently, the fit describes the data well. The good fit supports the assumption that the enhancement factor originates from enhanced surface mobility. From the fit, we obtain $\alpha = 3 \pm 0.5$ and $\xi_0 = (7.6 \pm 1) \times 10^{17} \text{ N s m}^{-3}$. Figure 5 displays the friction reduction factor, ξ/ξ_0 , as a function of film thickness h_0 according to the fitted result.

As far as we know, surface-promoted friction reduction has not been observed before. To explore this phenomenon further, we study the h_0 dependence of η_{eff} for three M_w 's in the plateau regime, namely, 115, 450, and 940 K g/mol. The result is shown in Figure 6. As the data show, η_{eff} maintains the same small- h_0 trend (i.e., the empirical $\sim h_0^3$ dependence seen in Figure 2b) only for $h_0/R_g < \sim 1$. Near $h_0/R_g \approx 1$, it jumps toward the bulk viscosity. A much slower rise, extending into $h_0/R_g > \sim 20$, was found in the dewetting of PS films on a liquid

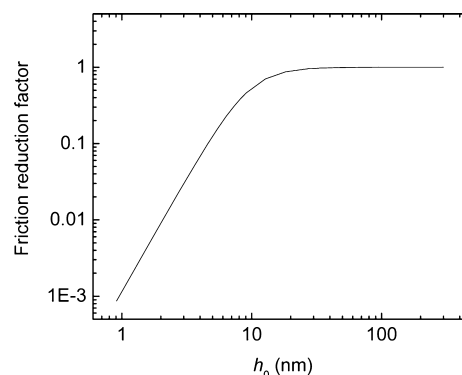


Figure 5. Friction reduction factor, ξ/ξ_0 , as a function of film thickness h_0 according to the fitted values of α and b_{mobile} found.

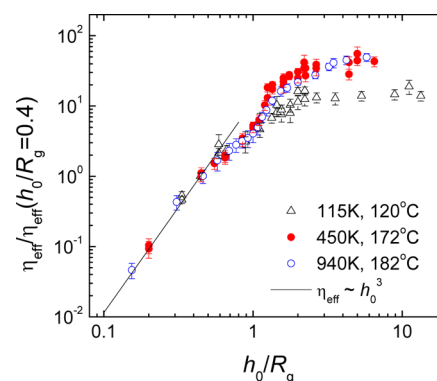


Figure 6. Normalized η_{eff} versus h_0/R_g of PS-SiOx with M_w and temperatures as specified in the legend. The solid line, showing the $\eta_{\text{eff}} \sim h_0^3$ scaling, guides the eye.

substrate,⁴⁷ where dominance of slippage is expected. It thus appears that slippage in our films becomes abruptly suppressed when h_0/R_g exceeds ~ 1 . But this is what one expects if the noted friction reduction is enabled by chain connectivity to the free surface.

The relation $M_{\text{tot}} \sim h_0^0$ found above means that thinner films transport the film fluid as efficiently as thicker films do. This has been identified as a signature of surface transport for unentangled films at lower temperatures close to T_g .⁸ As discussed above, surface transport is ruled out in this case because the much longer chains residing simultaneously in both layers forbid sustained surface flow.

In fitting the data to eq 1, we also deduce the shear modulus μ_0 . We consistently find that $\mu_0 \sim 10$ kPa or $\sim 1/10$ times the bulk value. A smaller modulus can be caused by reduced molecular coordination or chain entanglement at the free surface³⁸ or reduced chain entanglement resulting from the film preparation process.³² Noting that μ_0 is extracted using eq 1 which assumes no slippage in the fast elastic vibrations. Imperfect stickiness of the substrate may also cause an apparent reduction in μ_0 . Previous experiments had found stiffening of ultrathin polymer films in the rubbery response regime.⁴⁸ Existence of slippage would explain why stiffening is not observed in our films.

The saturation of η_{eff} at large M_w roughly occurs when R_g becomes $\sim h_0$. However, we believe that this is a coincidence. The criterion $R_g \sim h_0$ would mean that the M_w at the crossover should scale as h_0^2 or occur much sooner at 12.5 and 34.8 kg/mol at $h_0 = 3$ and 5 nm, respectively. This clearly does

not agree with our data. Success of eq 4 in describing the data reveals that the saturation of η_{eff} results from competition between flow mechanisms, namely surface-enhanced shear flow (through enhanced surface mobility) and surface-promoted plug flow (through surface-promoted friction reduction).

Stratton⁴⁹ had found that the viscosity of PS became independent of M_w when the shear rate, $\dot{\gamma}$, was increased above the reptation rate, τ_{rep}^{-1} . This phenomenon was understood to be caused by convective constraint release^{50,51}—a condition that is created when the entanglement points are swept away by the flow of surrounding chains faster than the chains can independently reptate away. In our experiments slippage dominates, so shear rate should be vanishingly small and approaches zero at long wavelengths. We now show that even under the no-slip boundary condition $\dot{\gamma}$ is too small for the occurrence of convective constraint release. We estimate the values of $\dot{\gamma}$ in this case for the 5 and 20 nm films as follows. For an order-of-magnitude estimate, we write $\dot{\gamma} \sim \langle v \rangle / h_0$, where $\langle v \rangle$ is the average fluid current in the film. To estimate $\langle v \rangle$, we use $M_{\text{tot}} \nabla P = j = \langle v \rangle h_0$, where j is the unit-width fluid current and P is the local Laplace pressure with which $\nabla P \sim \gamma_s q^3 \delta h$ for surface undulations with wavevector q and amplitude δh . For δh , we approximate it by the experimental root-mean-square roughness of the films, which is <0.2 and <0.27 nm for the 5 and 20 nm films, respectively. For M_{tot} we use $M_{\text{tot}} \equiv h_0^3 / (3\eta_{\text{eff}})$ and the plateau values of η_{eff} . These give $\dot{\gamma} = 1.0 \times 10^{-6}$ and $1.8 \times 10^{-8} \text{ s}^{-1}$ for the 5 and 20 nm films, respectively. For the reptation rate, we crudely estimate it by the experimental value of $\mu_0 / \eta_{\text{eff}}$ which gives values ~ 1000 times those of $\dot{\gamma}$ found above. This shows that convective constraint release is improbable in this experiment.

In the same vein, the dynamics observed in this experiment should not be compared with those in dewetting experiments.^{31,51} In dewetting experiments, the stress (or pressure differential), σ , that drives holes opening is provided by the surface energy change upon holes enlargement which is $\sim \gamma_s / h_0$. For $h_0 = 5$ nm and $\gamma_s \approx 30$ mN/m, $\sigma \sim 6$ MPa. This is bigger than the rubbery elastic modulus of common polymers and so can cause plasticization and nonlinear viscoelastic dynamics. In our experiment, the stress is provided by the Laplace pressure differential (which may be approximated by that between the crest and trough of the small surface undulations assumed to be sinusoidal), $\sim \gamma_s q^2 \delta h = (\gamma_s / h_0)(qh_0)^2(\delta h / h_0)$. For the 5 nm (20 nm) films, $qh_0 = 0.05$ (0.2) and $\delta h / h_0 = 0.04$ (0.014). These values imply that the stress in either the 5 or 20 nm film is only about 1/10000 times that in dewetting experiments. Henceforth, while nonlinear viscoelastic dynamics are common in dewetting experiments, it is unlikely in ours.

Experiments of polymer diffusion involving no surface energy change had more often than not found that the effective diffusivity decreased with increasing degree of confinement.^{42,52} By assuming the friction coefficient near the confining wall to be higher than the bulk friction coefficient, and the observed diffusivity to be governed by the friction coefficient averaged over the sample volume, Lange et al.⁴³ were able to account for several experimental results. In particular, on applying their model to the data of Frank et al.,⁵² they deduced that the near-substrate friction coefficient of PS supported by silicon was enhanced by a factor of 155 relative to the polymer–polymer friction coefficient. In this experiment, the near-substrate friction coefficient is ξ . Our model analysis suggests that under the ultrathin condition $h_0 / R_g < 1$ ξ decreases with decreasing h_0 from the $h_0 \rightarrow \infty$ value (i.e., ξ_0) in a manner

displayed in Figure 5. By using the published monomeric friction coefficient of bulk PS at 169.5 °C,⁵³ $\xi_{\text{mono}}(169.5 \text{ °C}) = 3.02 \times 10^{-8} \text{ N s m}^{-1}$ or approximately $9.74 \times 10^{10} \text{ N s m}^{-3}$, and the assumption $\xi_{\text{mono}}(120 \text{ °C}) / \xi_{\text{mono}}(169.5 \text{ °C}) = \eta_{\text{bulk}}(120 \text{ °C}) / \eta_{\text{bulk}}(169.5 \text{ °C})$, we estimate $\xi_0 / \xi_{\text{mono}}$ to be ~ 600 , which is bigger than but in order-of-magnitude agreement with Lange et al.'s estimate. On the other hand, one may observe that the calculation of Lange et al. does not include any possible reduction in the friction coefficient in the near-free surface region, so can potentially underestimate the effect of the substrate. According to Figure 5, ξ becomes equal to $\xi_0 / 600$ or ξ_{mono} at approximately $h_0 = 1$ nm. Since viscosity plateau is clearly seen in this experiment for the films with $h_0 > 1$ nm, our result indicates that interfacial slippage may occur even when the polymer–substrate friction coefficient is bigger than the polymer–polymer friction coefficient. But this should be of no surprise. As noted above, interfacial slippage results from competition between flow mechanisms whereby plug flow dominates at high M_w but is superseded by enhanced surface flow at low M_w . The condition varies with diffusivity measurements of polymer under confinement. According to Lange et al., enlargement in the average friction coefficient due to enhanced polymer friction at the substrate should always cause the observed diffusivity to decrease, with the amount of diffusivity reduction being independent of M_w . The latter is in keeping with Lange et al.'s measurement that for unentangled to mildly entangled polybutadiene in AAO nano channels, the amount of diffusivity reduction only depended on the channel size but not M_w .⁴²

CONCLUSION

In this experiment, we measured film mobility of PS confined between a free surface and a solid substrate for various h_0 and M_w . For small M_w , film flow exhibits $\eta_{\text{eff}} \sim M_w^{3.2 \pm 0.2}$ dynamics with insignificant slippage at the substrate. The data can be modeled by a two-layer model, assuming the films to consist of a surface mobile layer atop a bulk-like layer. At high M_w , slippage dominates and the dynamics is dictated by polymer–substrate interfacial friction that decreases with decreasing h_0 . We attribute this confinement effect to surface-promoted friction reduction, the impact of which manages to dominate any surface-enhanced shear flow at large M_w .

ASSOCIATED CONTENT

Supporting Information

Brief information about the R_g data cited in the text and Figure S1. The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.macromol.5b01002.

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Notes

The authors declare no competing financial interest.

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